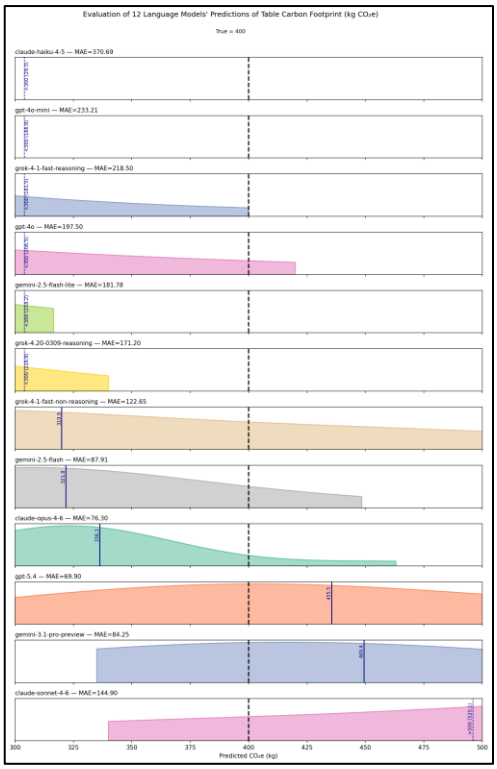
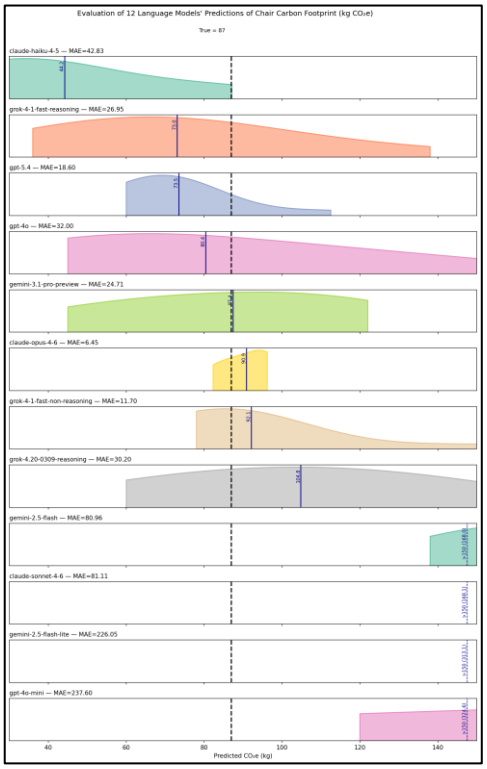
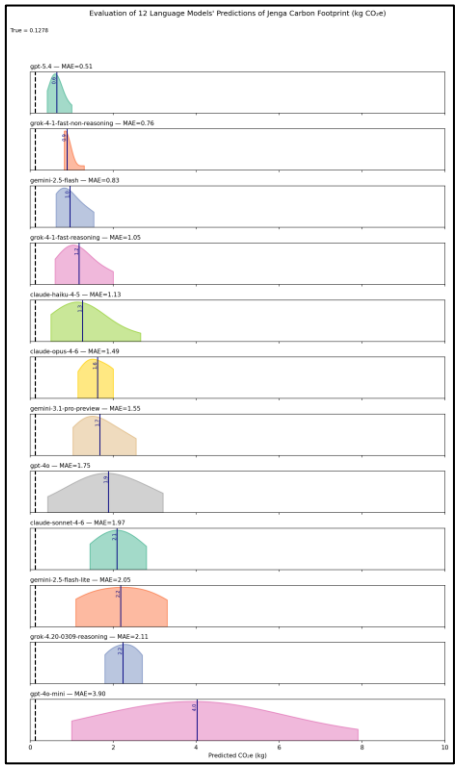
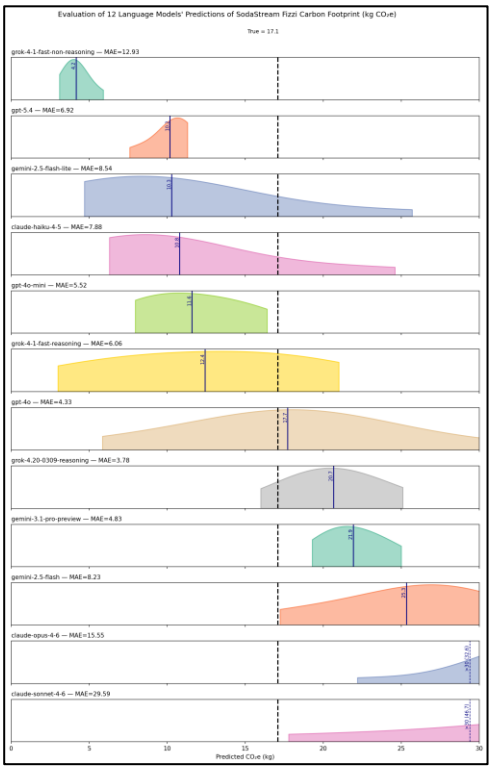
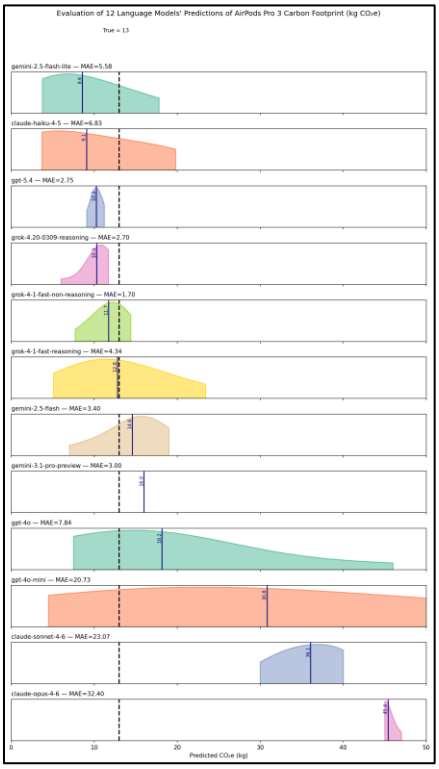
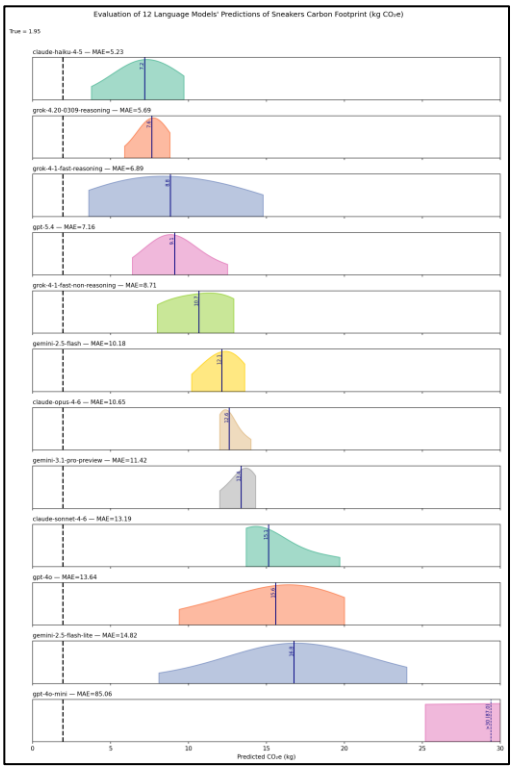

Evaluation of 12 Language Models' Predictions of Table Carbon Footprint (kg CO₂e)
True = 400
claude-haiku-4-5 — MAE=370.69
gpt-4o-mini — MAE=233.21
grok-4-1-fast-reasoning — MAE=218.50
gpt-4o — MAE=197.50
gemini-2.5-flash-lite — MAE=181.78
grok-4.20-0309-reasoning — MAE=171.20
grok-4-1-fast-non-reasoning — MAE=122.65
gemini-2.5-flash — MAE=87.91
claude-opus-4-6 — MAE=76.30
gpt-5.4 — MAE=69.90
gemini-3.1-pro-preview — MAE=84.25
claude-sonnet-4-6 — MAE=144.90
Predicted CO₂e (kg)


Evaluation of 12 Language Models' Predictions of Chair Carbon Footprint (kg CO₂e)
True = 87
claude-haiku-4-5 — MAE=42.83
grok-4-1-fast-reasoning — MAE=26.95
gpt-5.4 — MAE=18.60
gpt-4o — MAE=32.00
gemini-3.1-pro-preview — MAE=24.71
claude-opus-4-6 — MAE=6.45
grok-4-1-fast-non-reasoning — MAE=11.70
grok-4.20-0309-reasoning — MAE=30.20
gemini-2.5-flash — MAE=80.96
claude-sonnet-4-6 — MAE=81.11
gemini-2.5-flash-lite — MAE=226.05
gpt-4o-mini — MAE=237.60
Predicted CO₂e (kg)


Evaluation of 12 Language Models' Predictions of Jenga Carbon Footprint (kg CO₂e)
True = 0.1278
gpt-5.4 — MAE=0.51
grok-4-1-fast-non-reasoning — MAE=0.76
gemini-2.5-flash — MAE=0.83
grok-4-1-fast-reasoning — MAE=1.05
claude-haiku-4-5 — MAE=1.13
claude-opus-4-6 — MAE=1.49
gemini-3.1-pro-preview — MAE=1.55
gpt-4o — MAE=1.75
claude-sonnet-4-6 — MAE=1.97
gemini-2.5-flash-lite — MAE=2.05
grok-4.20-0309-reasoning — MAE=2.11
gpt-4o-mini — MAE=3.98
Predicted CO₂e (kg)


Evaluation of 12 Language Models' Predictions of SodaStream Fizzi Carbon Footprint (kg CO₂e)
True = 17.1
grok-4-1-fast-non-reasoning — MAE=12.93
gpt-5.4 — MAE=6.92
gemini-2.5-flash-lite — MAE=8.54
claude-haiku-4-5 — MAE=7.88
gpt-4o-mini — MAE=5.52
grok-4-1-fast-reasoning — MAE=6.06
gpt-4o — MAE=4.33
grok-4.20-0309-reasoning — MAE=3.78
gemini-3.1-pro-preview — MAE=4.83
gemini-2.5-flash — MAE=8.23
claude-opus-4-6 — MAE=15.55
claude-sonnet-4-6 — MAE=29.59
Predicted CO₂e (kg)


Evaluation of 12 Language Models' Predictions of AirPods Pro 3 Carbon Footprint (kg CO₂e)
True = 13
gemini-2.5-flash-lite — MAE=5.58
claude-haiku-4-5 — MAE=6.83
gpt-5.4 — MAE=2.75
grok-4.20-0309-reasoning — MAE=2.70
grok-4-1-fast-non-reasoning — MAE=1.70
grok-4-1-fast-reasoning — MAE=4.34
gemini-2.5-flash — MAE=3.40
gemini-3.1-pro-preview — MAE=3.00
gpt-4o — MAE=7.84
gpt-4o-mini — MAE=20.73
claude-sonnet-4-6 — MAE=23.07
claude-opus-4-6 — MAE=32.40
Predicted CO₂e (kg)


Evaluation of 12 Language Models' Predictions of Sneakers Carbon Footprint (kg CO₂e)
True = 1.95
claude-haiku-4-5 — MAE=5.23
grok-4.20-0309-reasoning — MAE=5.69
grok-4-1-fast-reasoning — MAE=6.89
gpt-5.4 — MAE=7.16
grok-4-1-fast-non-reasoning — MAE=8.71
gemini-2.5-flash — MAE=10.18
claude-opus-4-6 — MAE=10.65
gemini-3.1-pro-preview — MAE=11.42
claude-sonnet-4-6 — MAE=13.19
gpt-4o — MAE=13.64
gemini-2.5-flash-lite — MAE=14.82
gpt-4o-mini — MAE=85.06
Predicted CO₂e (kg)